

# WEEK 3 – SELLER & INVENTORY MANAGEMENT SYSTEM

## E-Commerce Database Management System

This document contains the table design, relationships, sample data, inventory analysis, and execution guidance for the Week 3 assignment.

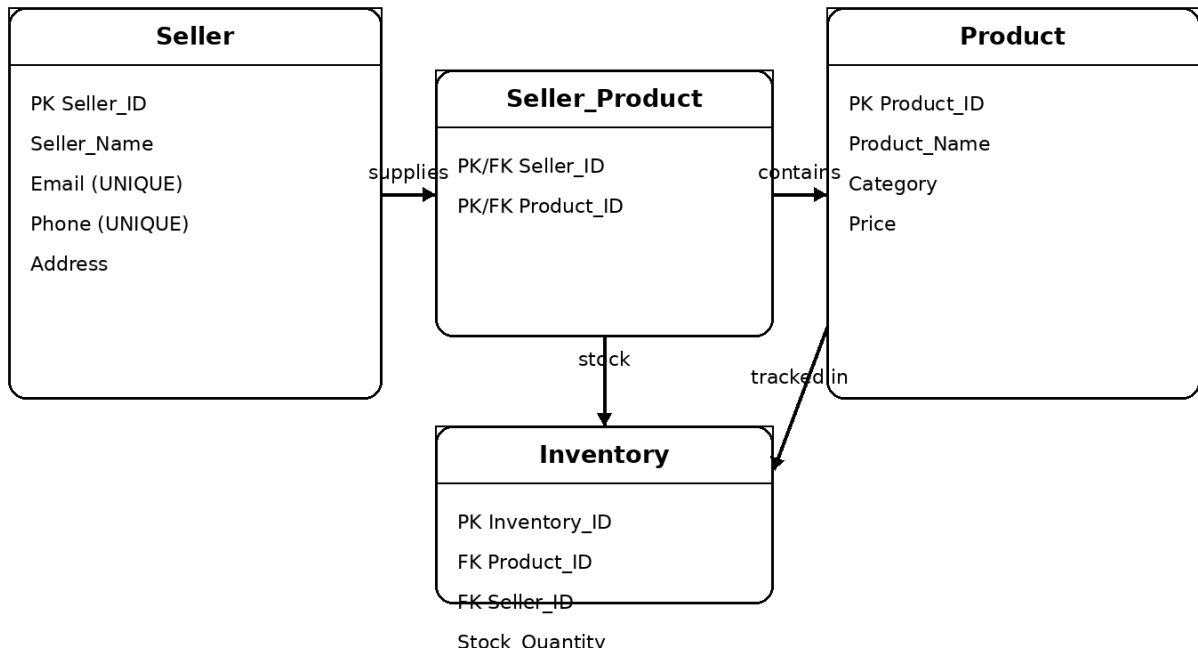
### 1. Seller & Inventory Table Design

Seller stores seller details. Product stores products. Seller\_Product connects sellers and products. Inventory stores stock quantity and status for a seller-product combination.

| Table          | Column         | Type          | Constraint           |
|----------------|----------------|---------------|----------------------|
| Seller         | Seller_ID      | INT           | PK, AUTO_INCREMENT   |
| Seller         | Seller_Name    | VARCHAR(100)  | NOT NULL, CHECK      |
| Seller         | Email          | VARCHAR(100)  | NOT NULL, UNIQUE     |
| Seller         | Phone          | VARCHAR(15)   | NOT NULL, UNIQUE     |
| Seller         | Address        | VARCHAR(255)  | NOT NULL             |
| Product        | Product_ID     | INT           | PK, AUTO_INCREMENT   |
| Product        | Product_Name   | VARCHAR(150)  | NOT NULL             |
| Product        | Category       | VARCHAR(100)  | NOT NULL             |
| Product        | Price          | DECIMAL(10,2) | NOT NULL, CHECK > 0  |
| Seller_Product | Seller_ID      | INT           | PK, FK               |
| Seller_Product | Product_ID     | INT           | PK, FK               |
| Inventory      | Inventory_ID   | INT           | PK, AUTO_INCREMENT   |
| Inventory      | Product_ID     | INT           | FK, NOT NULL         |
| Inventory      | Seller_ID      | INT           | FK, NOT NULL         |
| Inventory      | Stock_Quantity | INT           | NOT NULL, CHECK >= 0 |
| Inventory      | Stock_Status   | VARCHAR(20)   | NOT NULL, CHECK      |
| Inventory      | Last_Updated   | DATETIME      | DEFAULT/UPDATE       |

### 2. Updated ER Diagram

## Updated ER Diagram - Seller & Inventory Management



### 3. Sample Seller Data

| ID | Seller             | Email                      | Phone      | Address                |
|----|--------------------|----------------------------|------------|------------------------|
| 1  | TechWorld Supplies | techworld@gmail.com        | 9876500011 | Chennai, Tamil Nadu    |
| 2  | Smart Choice Store | smartchoice@gmail.com      | 9876500012 | Coimbatore, Tamil Nadu |
| 3  | Digital Hub        | digitalhub@gmail.com       | 9876500013 | Madurai, Tamil Nadu    |
| 4  | Prime Electronics  | primeelectronics@gmail.com | 9876500014 | Trichy, Tamil Nadu     |
| 5  | NextGen Traders    | nextgen@gmail.com          | 9876500015 | Salem, Tamil Nadu      |

### 4. Sample Product Data

| ID | Product                  | Category    | Price    |
|----|--------------------------|-------------|----------|
| 1  | HP Pavilion 15           | Laptop      | 58999.00 |
| 2  | Dell Inspiron 15         | Laptop      | 62999.00 |
| 3  | Lenovo IdeaPad Slim 3    | Laptop      | 47999.00 |
| 4  | Logitech Wireless Mouse  | Accessories | 1299.00  |
| 5  | HP USB Keyboard          | Accessories | 899.00   |
| 6  | Samsung 24 Inch Monitor  | Monitor     | 11999.00 |
| 7  | Canon Wireless Printer   | Printer     | 15499.00 |
| 8  | JBL Bluetooth Speaker    | Audio       | 3499.00  |
| 9  | TP-Link WiFi Router      | Networking  | 2499.00  |
| 10 | Seagate 1TB External HDD | Storage     | 5299.00  |

## 5. Sample Inventory Data

| ID | Product                  | Seller             | Stock | Status       |
|----|--------------------------|--------------------|-------|--------------|
| 1  | HP Pavilion 15           | TechWorld Supplies | 25    | Available    |
| 2  | Logitech Wireless Mouse  | TechWorld Supplies | 8     | Low Stock    |
| 3  | Samsung 24 Inch Monitor  | TechWorld Supplies | 15    | Available    |
| 4  | TP-Link WiFi Router      | TechWorld Supplies | 0     | Out of Stock |
| 5  | Dell Inspiron 15         | Smart Choice Store | 18    | Available    |
| 6  | HP USB Keyboard          | Smart Choice Store | 7     | Low Stock    |
| 7  | JBL Bluetooth Speaker    | Smart Choice Store | 0     | Out of Stock |
| 8  | Seagate 1TB External HDD | Smart Choice Store | 22    | Available    |
| 9  | Lenovo IdeaPad Slim 3    | Digital Hub        | 30    | Available    |
| 10 | Logitech Wireless Mouse  | Digital Hub        | 12    | Available    |
| 11 | Canon Wireless Printer   | Digital Hub        | 5     | Low Stock    |
| 12 | TP-Link WiFi Router      | Digital Hub        | 9     | Low Stock    |
| 13 | HP Pavilion 15           | Prime Electronics  | 14    | Available    |
| 14 | Samsung 24 Inch Monitor  | Prime Electronics  | 4     | Low Stock    |
| 15 | Canon Wireless Printer   | Prime Electronics  | 0     | Out of Stock |
| 16 | Seagate 1TB External HDD | Prime Electronics  | 16    | Available    |
| 17 | Dell Inspiron 15         | NextGen Traders    | 11    | Available    |
| 18 | Lenovo IdeaPad Slim 3    | NextGen Traders    | 6     | Low Stock    |
| 19 | HP USB Keyboard          | NextGen Traders    | 20    | Available    |
| 20 | JBL Bluetooth Speaker    | NextGen Traders    | 0     | Out of Stock |

## 6. Inventory Analysis Report

- Total inventory records: 20
- Total available units: 222
- Out-of-stock records: 4
- Low-stock records: 10
- Average inventory quantity: 11.10 units
- Highest stocked product record: Lenovo IdeaPad Slim 3 – Digital Hub (30 units)

## 7. Stock Availability Report

| Status       | Records | Units |
|--------------|---------|-------|
| Available    | 11      | 185   |
| Low Stock    | 5       | 39    |
| Out of Stock | 4       | 0     |

## 8. Execution Steps

Open MySQL Workbench or another MySQL 8.x client.

Open Week3\_Seller\_Inventory\_Management.sql.

Execute the script from the beginning. It creates SellerInventoryDB and all required tables/data.

Run the report queries at the end to generate inventory analysis.

Take screenshots of the query results if your instructor requires actual screenshots.

Upload the SQL, DOCX/PDF, and ER diagram to your GitHub repository.